

What Drives the Referees' Performance?

The Effect of Referee Coaches on Referees' Performance

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Abstract

Referees in any sport can substantially influence match outcomes. Therefore, the natural aim of every sports association is to maximise referee quality. The current literature lacks empirical evidence on the effects of various training methods, especially coaching referees. I estimate the effect of cumulative previous observations by a referee coach on the referee's received grade. Using a novel panel dataset with multiple controls and referee, coach, and referee-pair fixed effects, constructed from both publicly available and internal data from Czech Floorball, combined with one Regional Referee Committee dataset created from Referee Coach Reports, I find a statistically significant effect of previous observations with diminishing returns up to the fifth observation. The cumulative magnitude after the initial five observations improves (lowers) the grade by 0.24 grade points, reaching 30% of the Standard Deviation in the referee's received grade.

This represents a meaningful improvement in referee quality over a relatively short time, as it takes just five matches to enhance referees' skills in the same way. In comparison, the similar effect could be achieved with three to four years of actual refereeing experience. Moreover, theoretical and practical seminars yield no statistically significant effect.

Keywords

Referee Performance, Referee Coaches, Czech Floorball, Fixed Effects

Declaration of Generative AI Use

In order to uphold AI transparency in research, I declare that Claude Opus 4.5 (Anthropic) was used for translation of some files, and supportive and editorial assistance. All outputs were reviewed and verified by the author and he assumes full responsibility for the final manuscript.

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1 Introduction

Referees are an essential part of every sport, with their performance often affecting the match outcomes. This leads to the aim of training referees to make as few mistakes as possible. To achieve this, one must educate referees, which is costly. These costs are ultimately borne by the clubs and players, who pay to participate in competitions and are most affected by referees. Very often, personalised feedback from a referee coach is considered the most effective way to do so. To get the effect of referee coaches on their performance, I estimate how the number of previous observations by a referee coach affects the referee's received grade.

Using both publicly available data from the Czech Floorball and the internal database, and committee data from the same organisation, I build a panel of referee-match observations over two seasons, featuring multiple match-related and referee-specific characteristics. I use panel OLS with various fixed-effect specifications to estimate the relationship.

I find a statistically significant effect of previous observations on the received grade, with diminishing returns. The accumulated effect over the first five matches (the turning point is the fifth observation) corresponds to 30% of the standard deviation in the Referee's Received Grade, aligning with the existing coaching literature.

To get the same effect, a referee would have to gain three to four years of experience or attend multiple theoretical seminars, which do not systematically yield statistically significant results and whose effect on performance therefore remains unclear.

The paper fills the gap described by Cunningham et al. (2022) by empirically examining referee training and development systems, focusing not on elite but on mostly novice and early-stage career referees. It also expands the literature on coaching by bridging the topic to the sports context.

2 Literature Review

In recent years, the use of data analytics in sports has grown significantly to enhance performance and decision-making. However, there is a limited body of research that specifically addresses the performance of sports referees and the effectiveness of interventions aimed at improving their decision-making skills.

There is, however, extensive literature that analyses coaching and mentoring in various professional settings. Several meta-analyses found significant positive effects of enrolling in a coaching programme on both personal and organisational outcomes (Jones et al., 2016; Theeboom et al., 2014). This paper brings a new perspective from sports environment.

Some sport studies document systematic referee biases, particularly home-team favouritism, but does not examine training interventions. These findings are consistent across multiple European football leagues (Dawson et al., 2007; Dohmen & Sauermann, 2016; Garicano et al., 2005; Scoppa,

2008; Sutter & Kocher, 2004).

Another branch of literature finds that increased stress levels can lead to more errors in officiating. The size of the crowd further amplifies this effect. Additionally, referees with greater expertise show higher resilience to stress-induced errors. (Mascarenhas et al., 2009; Pizzera et al., 2022). A meta-study by Cunningham et al. (2022) reviews the findings of multiple studies that focus on the training and development of sports officials. They conclude that most of the studies lack retention measures, which makes it difficult to assess the long-term effectiveness of training programs. Furthermore, most current studies focus on elite referees during the advanced stages of their careers, leaving a gap in understanding the developmental needs of novice referees.

This is precisely where this paper aims to make its contribution. The unique dataset from one of the regions of the Czech Floorball allows an examination of the effect of various training methods on referee performance, while controlling for individual and match-specific characteristics. Furthermore, the data describe both novice and experienced referees at the regional level. I specifically focus on the role of referee coaches, who provide personalised feedback and guidance to referees after selected matches.

3 Czech Floorball & The Referee System

Floorball (also known as unihockey, innebandy or salibandy) is an indoor sport similar to ice hockey. It is played on a 40-by-20-meter pitch, with 5 field players and 1 goalkeeper per team. The match is typically divided into three 20-minute periods and is judged by 2 referees with equal authority (International Floorball Federation, 2022).

Czech Floorball is divided into seven regional associations and the national governing body. Each of them features a Referee Committee responsible for training, education, and referee nominations. The core of the education system, however, remains the same across all regions.

The educational system comprises multiple methods. First, the only obligatory requirement is the annual basic theoretical training, which referees must complete to be allowed to judge matches in a given season. Another way consists of theoretical and practical seminars that enable them to expand their formal education and obtain a higher license. Furthermore, there are matches alongside more experienced colleagues who serve as mentors, as well as a Special Development Programme for talented referees.

The final way of training is represented by the matches observed by a referee coach. This is often considered the most effective way to educate a referee and enhance their ability quickly. The referee coaches observe a match of a referee pair and provide them with feedback after it. They discuss and consult on their performance and "show them the right path". They also produce a report summarising their performance, leadership, and communication skills, as well as the application of the rules, and evaluate their performance by grade. (Czech Floorball, 2025a, 2025b, 2025c).

3.1 The Pairing Mechanism

Referees are paired based on their preferred colleague, roster assignment, and geographic distance to the match venue. Matching with preferred colleague does not happen automatically, as referees choose the days to judge independently, often not aligning with their colleagues’.

Five regional rosters cover different competition levels, with referees assigned twice per season. When availability is limited, referees judge outside their roster, making specific match assignments unpredictable. The data also allow me to control for match and competition difficulty (Czech Floorball, 2025a, 2025b).

Referee coaches are assigned to matches independently of the referee pairing mechanism, with the primary goal of observing as many different referees as possible. Logistical factors drive assignment; referee quality or recent performance do not systematically affect coach allocation (Czech Floorball, 2025a).¹

4 Data

I utilised multiple data sources and later matched them based on the referees’ names and birthdates. The data was later anonymised to protect the referees’ privacy.

Firstly, data on match statistics and referees’ participation in theoretical and practical seminars were obtained from publicly available online sources: match reports, seminar attendance lists, and specific competition rules.²

The second part of the data was acquired directly from the Czech Floorball’s internal information system. This data included more precise referees’ personal information (name, birthdate, license level, region, test errors, etc.). Finally, the remaining data were obtained from the Regional Referee Committee. This data included the referee’s years of experience, the coefficients for the difficulty of different competitions, and, most importantly, summaries of the referee coaches’ reports. These reports are written after each match observed by a referee coach and also feature a grade assigned to each referee separately, as well as a measure of the match’s difficulty. The English translation of the form used by referee coaches is provided in Appendix Figure B.3.³

The final dataset matches the 2023/24 and 2024/25 season data from the Hradec Králové and Pardubice region⁴ of the Czech Floorball. This region was selected because of data availability resulting from the report structure. The other regions differ in reporting norms. The education system of referees and the purpose of referee coaches are the same, the same goes for competitions. There are no known systematic differences in referee performance across regions.

¹This information is also based on the author’s firsthand experience as a member of the Regional Referee Committee.

²For an example match report, see Czech Floorball (2025d, 2025e), for an example of seminar entry, visit Czech Floorball (2025f).

³The translation was produced using Claude Opus 4.5 (Anthropic) and subsequently reviewed by a human.

⁴For the location of the regions, see Figure B.4 in the Appendix

Table 1: Summary Statistics for Selected Variables

	N	Mean	Std. Dev.	Min	Max
Referee's Received Grade	442	2.06	0.79	1.00	4.00
Previous Observations	442	2.30	2.31	0.00	12.00
Match Difficulty	442	2.02	0.65	1.00	4.00
Competition Difficulty	442	0.61	0.24	0.30	1.00
Completed Seminars	442	13.59	11.78	1.00	53.00
Referee Experience	442	4.53	4.19	1.00	19.00
Number of Penalties Awarded in a Match	442	4.72	5.36	0.00	30.00
Leave-One-Out Average Penalties in a Season	442	3.26	2.29	0.00	8.49
Permanent Colleague Indicator	442	0.33	0.47	0.00	1.00
Age	442	22.83	8.71	15.06	50.81

The resulting matched dataset contains 442 individual observations of 112 different referees judging 221 matches over two seasons. No data was lost during the matching process. Table 1 presents summary statistics for the main variables. Summary Statistics for all the variables can be found in Appendix Table A.1, for further explanation of all the variables, see Appendix Table A.2. The mean referee's grade is 2.06 on a scale from 1 (best) to 4 (worst) which indicates that, on average, referees "handled the match level without difficulty". Figure A.1 in the Appendix presents the overall distribution of the referee's received grades, showing sufficient variation in the dependent variable. Figure A.2 in the Appendix presents the correlation matrix heatmap for all the variables used in the analysis.

5 Empirical Strategy & Methodology

I test whether the number of cumulative previous matches observed and consulted by referee coaches has a significant effect on referee performance. I use the Referee's Received Grade as a proxy for performance, and examine potential nonlinear effects of cumulative observations.

The baseline linear specification is:

$$Y_{itcp} = \beta_1 \text{PrevObs}_{it} + X'_{it}\gamma + \eta + \varepsilon_{itcp}, \quad (1)$$

where Y_{itcp} is the grade received by referee i in match t from coach c while paired with partner p , and PrevObs_{it} counts cumulative prior observations. X_{it} is a vector of controls, and $\eta \in \{\delta_i, \delta_i + \theta_c, \rho_p + \theta_c\}$ represents fixed effects combinations: δ_i captures time-invariant referee ability, θ_c accounts for coach-specific grading severity, and ρ_p controls for pair-specific characteristics.

The quadratic specification incorporates squared terms to capture potential nonlinearities:

$$Y_{itcp} = \beta_1 \text{PrevObs}_{it} + \beta_2 \text{PrevObs}_{it}^2 + \beta_3 \text{Exp}_{it} + \beta_4 \text{Exp}_{it}^2 + X'_{it}\gamma + \eta + \varepsilon_{itcp}, \quad (2)$$

where Exp_{it} denotes years of refereeing experience.

The vector of controls X_{it} includes different combinations of variables related to referee and match characteristics (competition difficulty, match difficulty, permanent colleague indicator, completed seminars, number of days offered for judging, leave-one-out average penalties in a season). The exact combination of controls varies across specifications to test robustness.

I estimate the equations using the Ordinary Least Squares (OLS) with standard errors clustered at the referee level to account for within-referee correlation over time. Identification relies on within-referee variation in cumulative observations over time. The key assumption is that, conditional on referee fixed effects and controls, the timing and frequency of coach observations is uncorrelated with unobserved determinants of performance.

If referee coaching is effective, I expect $\beta_1 < 0$, indicating that additional observations lead to lower (better) grades. The quadratic specification allows for diminishing returns: a negative β_1 combined with a positive β_2 would imply that early observations improve performance (lower grades), but the marginal effect diminishes with additional observations. The turning point occurs at $-\frac{\beta_1}{2\beta_2}$ observations. Singleton observations of referees are dropped from all fixed effects specifications.

6 Results

Table 2 presents the main results from estimating the effect of previous observations on referee performance. The first column shows the results from the model with basic controls, together with referee fixed effects and coach fixed effects. The second column adds another set of controls. Columns three and four replicate this structure but use referee pair fixed effects instead of Referee fixed effects.

The coefficient on previous observations is negative and statistically significant across all specifications, indicating that each additional observation by a referee coach is associated with an improvement in referee performance (lower grades). The magnitude of the effect varies slightly across specifications but remains consistent in sign and significance. The quadratic specifications reveal a positive coefficient on the squared term, suggesting diminishing returns to coaching observations. The fourth column of Table 2 indicates that the first observations yield the most noticeable improvements, with the effect after the first observation being -0.092 grade points. This small effect continues to diminish until it reaches the turning point at 4.68 previous observations. Column three indicated the turning point at exactly 5 previous observations.

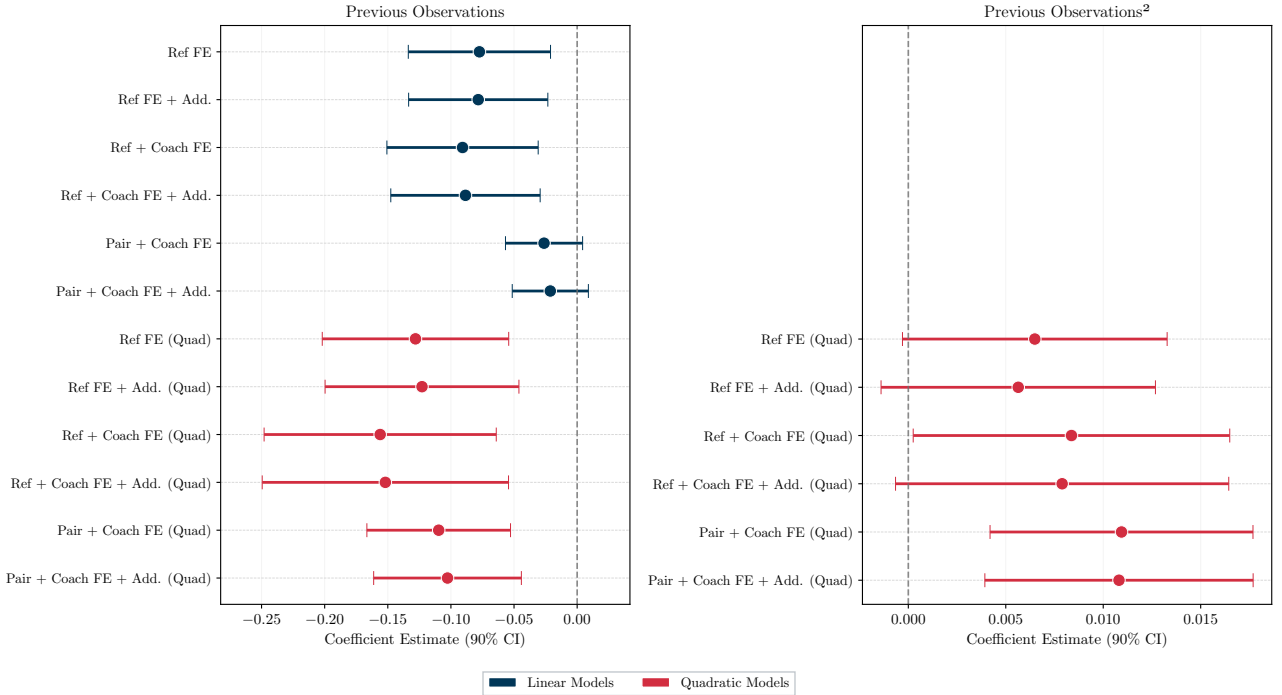
In contrast, the coefficient on referee experience is not consistent in sign across specifications. However, once referee pair fixed effects are included, the sign becomes negative, aligning with expectations that experience should improve performance. The quadratic term for experience is positive in last two specifications, indicating diminishing returns to experience up to the turning point after 12 to 14 years. Column two also highlights the effect of pairing with a preferred colleague, which is associated with significantly better performance.

Table 2: Effect of Referee Coaches on Referee Performance

	Referee's Received Grade			
	(1)	(2)	(3)	(4)
Previous Observations	-0.156*** (0.056)	-0.152** (0.059)	-0.110*** (0.034)	-0.103*** (0.035)
Previous Observations ²	0.008* (0.005)	0.008 (0.005)	0.011*** (0.004)	0.011** (0.004)
Completed Seminars	0.009 (0.027)	0.004 (0.028)	-0.010** (0.005)	-0.004 (0.006)
Referee Experience	0.460** (0.226)	0.480* (0.258)	-0.076** (0.032)	-0.085** (0.035)
Referee Experience ²	-0.010 (0.008)	-0.009 (0.010)	0.003** (0.001)	0.003** (0.002)
Permanent Colleague Indicator		-0.252*** (0.092)		
Coach FE	✓	✓	✓	✓
Referee Pair FE	-	-	✓	✓
Referee FE	✓	✓	-	-
Basic Controls	✓	✓	✓	✓
Additional Controls	-	✓	-	✓
Observations	431	431	439	439
R^2	0.570	0.584	0.814	0.819
R^2 Within	0.219	0.243	0.237	0.260

Notes: Standard errors clustered at the referee level in parentheses. The dependent variable is the grade assigned by a referee coach after the observed match (1 = Excellent, 2 = Very Good, 3 = Sufficient, 4 = Insufficient; lower is better). Previous Observations counts prior matches in which the referee was evaluated in the dataset. Referee Experience measures years of refereeing (1 = first season). Permanent Colleague Indicator equals one if judging with a preferred colleague. It appears only in column (2) as it is dropped from column (4) because of multicollinearity. Basic Controls include Match Difficulty, and Competition Difficulty. Additional Controls consist of Leave-one-out Average Penalties, and Number of Days Offered for Judging. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure 1: Coefficient Plot of Previous Observations Effect



For expanded results with different combinations of controls and fixed effects, see Appendix Tables A.3 and A.4. The results remain robust across various model specifications, consistently showing

that referee coaching positively impacts performance, as shown in Figure 1.

7 Discussion

The results demonstrate the positive impact of referee coaching on performance, with additional observations by coaches leading to improvements in referees' grades. This finding aligns with the broader literature on coaching effectiveness in professional settings (Jones et al., 2016; Theeboom et al., 2014), extending its applicability to the sports officiating context - a context where Cunningham et al. (2022) identified a gap in the studies on novice referee development.

After the first five observations, the *ceteris paribus* cumulative effect of previous observations by a referee coach on referees' performance, according to column 4 of Table 2, sums up to 0.24 grade points on average, which equals 30% of the referee's received grade standard deviation, similar to Jones et al. (2016), who identified the coaching effect to be 36% of standard deviation in the dependent variable for skill-based outcomes in workplace coaching.

The magnitude of the effect might appear modest, however, its strength lies in its time-effectiveness. Each observation requires only one match to be judged and coached. If I consider the impact of referee experience from the same specification, the first year of experience improves the grade by 0.082 points, on average. To achieve the same cumulative effect as five coaching observations, a referee would need to gain three to four years of experience.

The first 2 columns of Table 2 show a positive sign for the experience. This could be happening due to several reasons. Firstly, coaches might hold experienced referees to higher standards, or referees might be assigned to more difficult matches, which the controls do not fully capture. Alternatively, the referees who stay in regional competitions and don't advance to the national ones might be those who are struggling to advance.

In contrast, completed seminars show no consistent significant effect on performance. This suggests that the personalised feedback and approach of referee coaching may be more effective than general theoretical training in improving referees' decision-making skills. The only significant coefficient (column 2, Table 2) has a magnitude of merely one-eighth of the effect of the initial coaching observation. Additionally, judging with a preferred colleague improves grades by 0.25 points—equivalent to roughly three coaching observations. This highlights the importance of pair stability in referee development.

Even though referee coach feedback is effective, it is threatened by the Hawthorne effect, as referees may alter their short-term behaviour and judging style to receive a better grade and be seen as a better referee, potentially biasing the distribution of grades. Nonetheless, the cumulative improvement combined with diminishing returns suggest that referees truly learn from the feedback over time rather than adjusting their behaviour temporarily.

Another concern lies in selection - if coaches observe referees who are already improving or more motivated, the estimated effect may be biased upwards. However, the assignment of referee

coaches is independent of referee quality or recent performance (Czech Floorball, 2025a)⁵, which mitigates this concern. Furthermore, the falsification test shown in Table A.5 provides evidence that future observations don't predict initial grades, suggesting coaches aren't assigned to already-better referees.

Nevertheless, the unbalanced panel structure, with some referees having more observations than others, could introduce bias if the frequency of observations correlates with unobserved referee characteristics. Future work with additional seasons of data could help address this issue.

The identification also relies on the assumption that the timing of coach observations is as good as random, conditional on controls and used fixed effects. It is further threatened by the fact that the coaches providing feedback assign grades themselves. There might be an incentive to evaluate the referees they see more often differently. While multiple coaches observe 53% of referees in the sample, 47% interact with only one coach. The fixed effects capture the systematic differences in grading but cannot account for this coach-specific favouritism. However, the diminishing returns from coaching suggest a pattern more consistent with learning than with favouritism. Future work could address this by utilising some objective measure of the performance or independent evaluations.

Causal identification could be strengthened with alternative research designs. If there were explicit scoring thresholds that led to different observation priorities, a regression discontinuity could compare referees just above and below the cut-offs. Alternatively, a future randomised experiment intentionally assigning the observation timing within cohorts of similar referees would provide the cleanest identification. Currently, the assignment is not purely random. However, it approximates randomness in practice.

The analysis also covers only one of seven Czech Floorball regions. While the educational system is standardised nationally, replication in other regions would strengthen generalisability, too. However, this is not possible as different regions use different Referee Coach Reports.

8 Conclusion

Using publicly available data and internal records from the Czech Floorball and its committees, I constructed a unique dataset that tracks referee performance over two seasons and features multiple match-related and referee-specific variables. This dataset allows for an empirical investigation into the effectiveness of referee coaching and other factors on referees' performance. I employed fixed-effects OLS models to estimate the effect of prior referee coach observations on referees' performance, measured by their received grades. The models control for various other factors, such as match difficulty, competition level, and pairing with preferred colleagues, and utilise multiple combinations of fixed effects to account for individual and pair characteristics.

⁵This information is also based on the author's firsthand experience as a member of the Regional Referee Committee.

The results show statistically significant and meaningful improvements in performance resulting from referee coaching. The cumulative effect of the first couple of coaching sessions is equivalent to several years of field experience, highlighting the efficiency of coaching as a training and development tool. The findings also align with the existing literature on coaching in professional settings, thereby extending its relevance to sports judging.

The paper opens the door for further research, including future revisiting of the analysis with datasets covering additional seasons, other regions of the Czech Floorball, or different sports altogether. Furthermore, analysing the patterns behind the fixed effects could bring insights into individual referee development. Overall, the study contributes to the understanding of referee development and the practical implications of coaching in sports officiating.

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Appendices

Appendix A Main Tables and Figures

Table A.1: Summary Statistics for All Variables

	N	Mean	Std. Dev.	Min	Max
Referee's Received Grade	442	2.06	0.79	1.00	4.00
Previous Observations	442	2.30	2.31	0.00	12.00
License Level	442	0.91	0.89	0.00	3.00
Competition Difficulty	442	0.61	0.24	0.30	1.00
Match Difficulty	442	2.02	0.65	1.00	4.00
Completed Seminars	442	13.59	11.78	1.00	53.00
Referee Experience	442	4.53	4.19	1.00	19.00
Permanent Colleague Indicator	442	0.33	0.47	0.00	1.00
Age	442	22.83	8.71	15.06	50.81
Number of Days Offered for Judging	442	30.99	15.17	7.00	75.00
Mistakes in the Season-Entry Test	442	3.12	2.24	0.00	11.00
New Referee Indicator	442	0.12	0.33	0.00	1.00
Number of Penalties Awarded in a Match	442	4.72	5.36	0.00	30.00
Leave-One-Out Average Penalties in a Season	442	3.26	2.29	0.00	8.49
Shots	442	41.00	21.61	0.00	113.00
Goals	442	9.76	4.57	1.00	31.00
Goal Difference	442	3.82	3.19	0.00	17.00
Shots Difference	442	8.05	8.14	0.00	35.00
Number of Visitors	442	37.57	67.09	0.00	421.00
Number of Visitors (Imputed)	442	42.97	65.21	0.00	421.00
Net Duration of the Match	442	43.02	12.49	20.00	60.00
Duration of the Match	442	74.19	28.31	20.00	163.00
Wage per Match	442	432.08	224.85	170.00	900.00
Future Observations	112	2.95	2.56	0.00	12.00

Table A.2: Explanation of All Variables

Variable	Description
Referee's Received Grade	Grade assigned by a coach in a match (1 = Excellent (referee performance significantly exceeded match level), 2 = Very Good (referees handled the match level without difficulty), 3 = Sufficient (referees slightly fell short of match level), 4 = Insufficient (referees significantly fell short of match level)).
Previous Observations	Number of prior matches in which the referee was evaluated.
License Level	Referee's license level (0–4, with 4 being the highest).
Competition Difficulty	Difficulty rating of the competition level (0.3–1.0, with 1.0 being the most difficult).
Completed Seminars	Number of theoretical and practical seminars completed by the referee until the observed match.
Match Difficulty	Difficulty level of the specific match (1–4, 1 = Easy, 2 = Standard Match for a Competition, 3 = Difficult, 4 = Very Difficult).
Permanent Colleague Indicator	Indicator for judging the match with a preferred colleague (1 = yes).
Age	Referee's age in years.
Referee Experience	Years of refereeing experience (1 = first season).
Number of Days Offered for Judging	Number of days the referee was available for judging.
Mistakes in the Season-Entry Test	Number of incorrect answers on the first attempt of the season-entry test.
New Referee Indicator	Indicator for first-season referees (1 = yes).
Number of Penalties Awarded in a Match	Number of penalties awarded in the match.
Leave-One-Out Average Penalties in a Season	Average number of penalties awarded by the referee in the season, excluding the current match.
Shots	Total number of shots in the match.
Goals	Total number of goals scored in the match.
Goal Difference	Goal difference between the teams in the match.
Shot Difference	Difference in total shots between the teams in the match.
Number of Visitors	Number of spectators attending the match.
Number of Visitors (Imputed)	Imputed number of spectators (used when unavailable; imputed from earlier matches in the same competition and sports hall on the same day).
Net Duration of the Match	Net duration of the match (in minutes).
Duration of the Match	Total duration of the match (in minutes).
Wage per Match	Wage received by the referee per match (in CZK).
Future Observations	The number of future observations after the initial match.

Figure A.1: Distribution of Referee's Received Grades

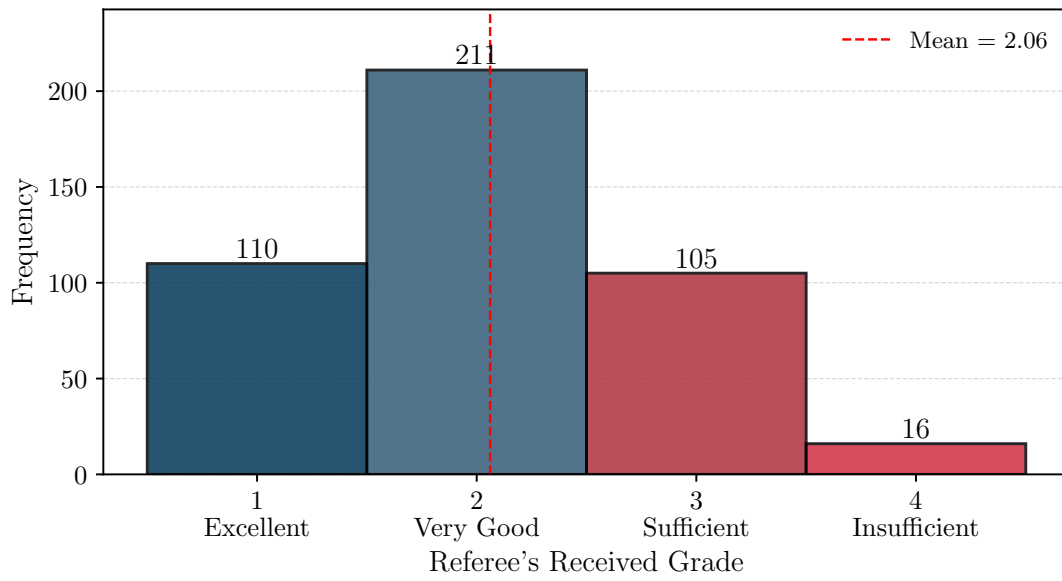


Figure A.2: Correlation Matrix Heatmap

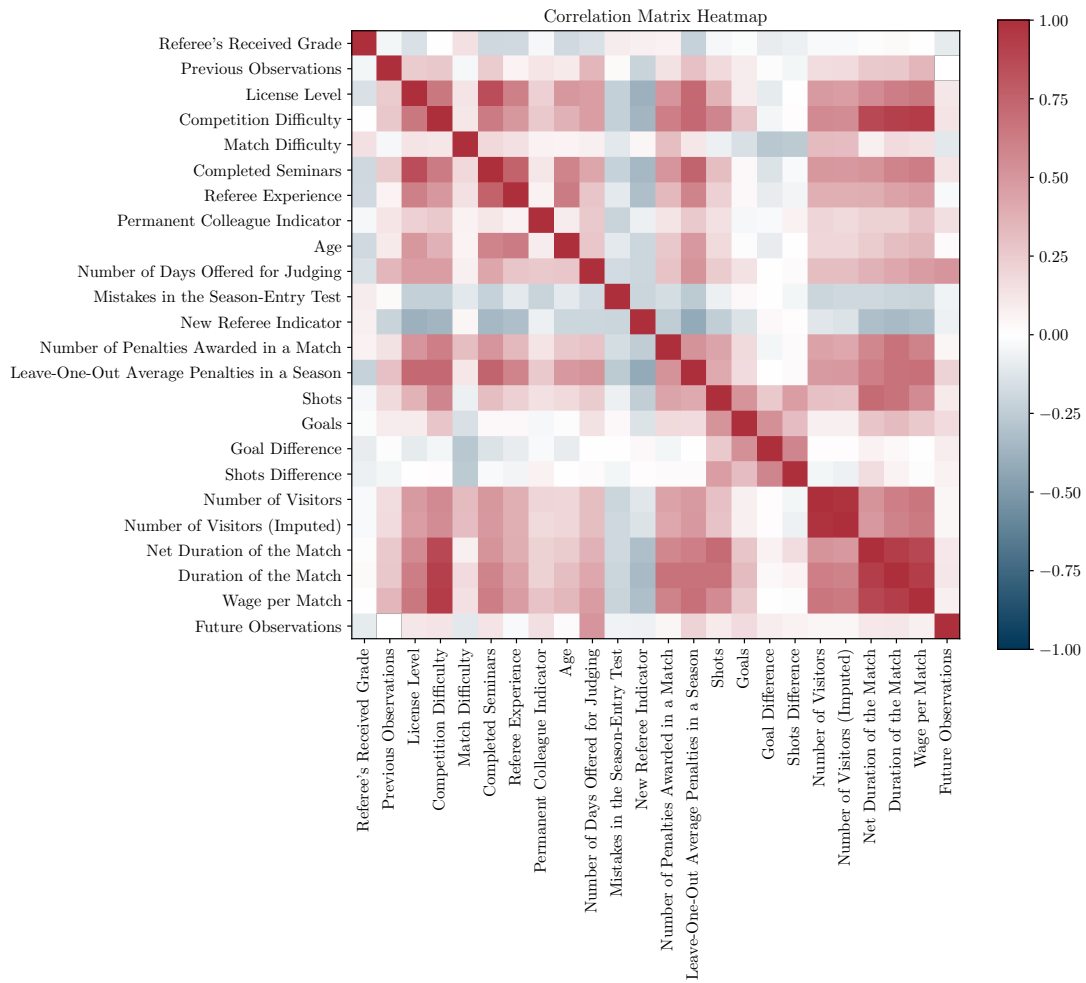


Table A.3: Effect of Referee Coaches on Referee Performance - Linear Models

	Referee's Received Grade					
	(1)	(2)	(3)	(4)	(5)	(6)
Previous Observations	-0.077** (0.034)	-0.078** (0.033)	-0.091** (0.036)	-0.088** (0.036)	-0.026 (0.019)	-0.021 (0.018)
Match Difficulty	0.315*** (0.068)	0.317*** (0.069)	0.335*** (0.071)	0.344*** (0.070)	0.404*** (0.089)	0.404*** (0.088)
Competition Difficulty	2.040*** (0.248)	2.103*** (0.247)	1.813*** (0.253)	1.915*** (0.219)	1.531*** (0.518)	1.527*** (0.516)
Completed Seminars	0.009 (0.025)	0.007 (0.026)	0.014 (0.027)	0.006 (0.029)	-0.013*** (0.005)	-0.008 (0.006)
Referee Experience	0.237 (0.161)	0.263 (0.169)	0.308* (0.182)	0.320* (0.176)	-0.019* (0.011)	-0.017 (0.011)
Leave-One-Out Average Penalties in a Season		-0.008 (0.097)		0.065 (0.096)		-0.071** (0.032)
Permanent Colleague Indicator		-0.174* (0.096)		-0.253*** (0.090)		
Number of Days Offered for Judging		-0.011 (0.013)		-0.016 (0.012)		-0.001 (0.002)
Coach FE	-	-	✓	✓	✓	✓
Referee Pair FE	-	-	-	-	✓	✓
Referee FE	✓	✓	✓	✓	-	-
Observations	431	431	431	431	439	439
R^2	0.527	0.534	0.564	0.579	0.806	0.811
R^2 Within	0.209	0.221	0.208	0.234	0.206	0.228

Notes: Standard errors clustered at the referee level in parentheses. The dependent variable is the grade assigned by a referee coach after the observed match (1 = Excellent, 2 = Very Good, 3 = Sufficient, 4 = Insufficient; lower is better). Previous Observations counts prior matches in which the referee was evaluated in the dataset. Match Difficulty rates the specific match (1 = Easy to 4 = Very Difficult). Competition Difficulty rates the competition level (0.3–1.0, higher = more difficult). Completed Seminars counts theoretical and practical seminars completed in the referee's lifetime. Referee Experience measures years of refereeing (1 = first season). Leave-One-Out Average Penalties is the referee's season average excluding the current match. Permanent Colleague Indicator equals one if judging with a preferred colleague. Number of Days Offered for Judging counts days the referee was available for judging. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A.4: Effect of Referee Coaches on Referee Performance - Quadratic Models

	Referee's Received Grade					
	(1)	(2)	(3)	(4)	(5)	(6)
Previous Observations	-0.128*** (0.045)	-0.123*** (0.047)	-0.156*** (0.056)	-0.152** (0.059)	-0.110*** (0.034)	-0.103*** (0.035)
Previous Observations ²	0.006 (0.004)	0.006 (0.004)	0.008* (0.005)	0.008 (0.005)	0.011*** (0.004)	0.011** (0.004)
Match Difficulty	0.313*** (0.069)	0.315*** (0.070)	0.333*** (0.070)	0.339*** (0.069)	0.395*** (0.081)	0.395*** (0.080)
Competition Difficulty	2.009*** (0.249)	2.070*** (0.250)	1.742*** (0.249)	1.846*** (0.220)	1.482*** (0.481)	1.466*** (0.477)
Completed Seminars	0.007 (0.025)	0.007 (0.026)	0.009 (0.027)	0.004 (0.028)	-0.010** (0.005)	-0.004 (0.006)
Referee Experience	0.307 (0.210)	0.359 (0.280)	0.460** (0.226)	0.480* (0.258)	-0.076** (0.032)	-0.085** (0.035)
Referee Experience ²	-0.004 (0.007)	-0.006 (0.011)	-0.010 (0.008)	-0.009 (0.010)	0.003** (0.001)	0.003** (0.002)
Leave-One-Out Average Penalties in a Season		-0.031 (0.129)		0.025 (0.114)		-0.069** (0.032)
Permanent Colleague Indicator		-0.170* (0.098)		-0.252*** (0.092)		
Number of Days Offered for Judging		-0.010 (0.012)		-0.015 (0.012)		-0.002 (0.002)
Coach FE	-	-	✓	✓	✓	✓
Referee Pair FE	-	-	-	-	✓	✓
Referee FE	✓	✓	✓	✓	-	-
Observations	431	431	431	431	439	439
R^2	0.530	0.537	0.570	0.584	0.814	0.819
R^2 Within	0.214	0.225	0.219	0.243	0.237	0.260

Notes: Standard errors clustered at the referee level in parentheses. The dependent variable is the grade assigned by a referee coach after the observed match (1 = Excellent, 2 = Very Good, 3 = Sufficient, 4 = Insufficient; lower is better). Previous Observations counts prior matches in which the referee was evaluated in the dataset. Match Difficulty rates the specific match (1 = Easy to 4 = Very Difficult). Competition Difficulty rates the competition level (0.3–1.0, higher = more difficult). Completed Seminars counts theoretical and practical seminars completed in the referee's lifetime. Referee Experience measures years of refereeing (1 = first season). Leave-One-Out Average Penalties is the referee's season average excluding the current match. Permanent Colleague Indicator equals one if judging with a preferred colleague. Number of Days Offered for Judging counts days the referee was available for judging. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A.5: Falsification Test: Future Observations and Initial Performance

	Referee's Received Grade			
	(1)	(2)	(3)	(4)
Future Observations	-0.030 (0.027)	-0.024 (0.032)	-0.041 (0.027)	-0.042 (0.035)
Match Difficulty	0.081 (0.129)	0.064 (0.126)	-0.038 (0.153)	-0.052 (0.153)
Competition Difficulty	0.077 (0.519)	0.643 (0.577)	-0.397 (0.585)	0.054 (0.723)
Completed Seminars	-0.003 (0.013)	0.007 (0.013)	0.010 (0.012)	0.018 (0.012)
Referee Experience	-0.032 (0.023)	-0.017 (0.023)	-0.059*** (0.021)	-0.045* (0.023)
Leave-One-Out Average Penalties in a Season		-0.153** (0.065)		-0.126** (0.063)
Permanent Colleague Indicator		0.050 (0.188)		0.117 (0.203)
Number of Days Offered for Judging		0.003 (0.006)		0.004 (0.007)
Intercept	2.191*** (0.376)	2.036*** (0.387)		
Coach FE	-	-	✓	✓
Observations	112	112	111	111
R^2	0.046	0.100	0.241	0.280
R^2 Within	-	-	0.097	0.144

Notes: Heteroskedasticity-robust standard errors in parentheses. Sample restricted to each referee's first observed match. The dependent variable is the grade assigned by a referee coach after the observed match (1 = Excellent, 2 = Very Good, 3 = Sufficient, 4 = Insufficient; lower is better). Future Observations counts subsequent coaching observations the referee will receive after this first match. Referee Experience measures years of refereeing (1 = first season). Permanent Colleague Indicator equals one if judging with a preferred colleague. Basic Controls include Match Difficulty, and Competition Difficulty. Additional Controls consist of Leave-one-out Average Penalties, and Number of Days Offered for Judging. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Appendix B Supplementary Materials

Figure B.3: Sample Referee Coach Report Translated to English

REFeree COACH REPORT - REFereES						 ČESKÝ FLORBAL	
Czech Floorball HKaPA version 2025							
Referee 1				Referee Roster		Competition	
Referee 2				Referee Roster		Date	
Home					R1 Readiness		
Away					R2 Readiness		
Referee Coach						R1	R2
	1st Period	2nd Period	3rd Period	Full Match	MATCH DIFFICULTY		
Goals Scored	:	:	:	:	SELF-ASSESSMENT		
Penalties by Period	:	:	:	:	GRADE		
Tolerance Level (R1/R2)	/	/	/	/			
Identity Check							
Match Report and Court Ch							
Referee Equipment:							
Jerseys				Shorts		Socks	
Brief Match Description (pace of play, match intensity, challenge intensity, technical and tactical quality and maturity of teams, emotional involvement of participants, key moments, referee approach):							
1. Match Management					Incorrect	Correct	Did Not Occur
Understanding and Feel for the Game							
Predictability and Consistency in Match Management							
Quick, Clear and Reliable Decisions							
Appropriate Reactions to Changes in Match Character							
Game Flow							
Advantages (Accuracy and Sensitivity in Awarding and Returning)							
Balance Between Referees							
Signalling - Technique, Clarity, Visibility, Timeliness							
Whistle Communication							
2. Personality					Incorrect	Correct	Did Not Occur
Match Preparation, Cooperation with Secretariat							
Authority, Charisma, Appearance, Acceptance, Facial Expression, Body Language							
Decisiveness, Handling Critical Situations							
Objectivity in Decisions, Ability to Resist Influence, Courage in Making Unpopular Decisions							
Communication - Prevention							
Communication - During and After Situations							
3. Rules Application					Incorrect	Correct	Did Not Occur
Physical Play Assessment Level (incl. Setting and Maintaining Standard)							
Stick Play Assessment Level (incl. Setting and Maintaining Standard)							
Appropriateness of Penalties Awarded							
Penalties Clearly Indicated, Penalized Players Correctly Identified							
Communication with Players After Penalties (Verbal and Non-verbal)							
Assessment and Penalization of Technical Offences							

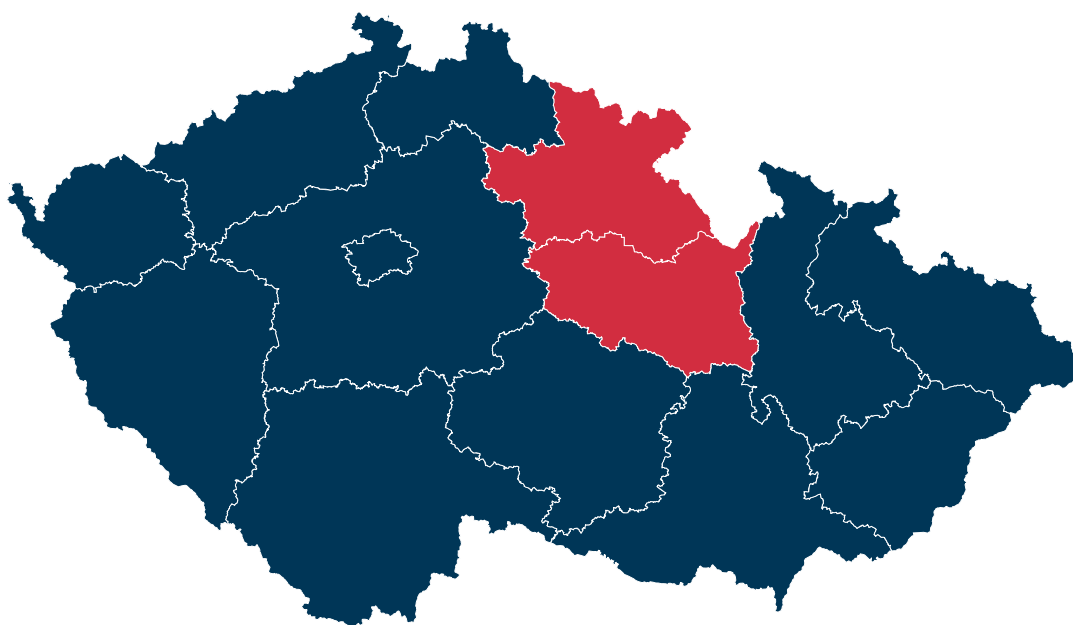
Assessment and Penalization of Unsporting Behaviour				
4. Positioning System		Incorrect	Correct	Did Not Occur
Goal Line, Goal Crease and Player Control				
Court Division, Side Changes				
Distance to Play, Position Selection, Closing Down Play				
Distance from Boards, Positioning Outside Corners				
Movement Behind Goal				
Positioning During Stoppages				
Communication Between Referees				
Overall Commentary and Performance Evaluation Evaluation of Individual Aspects (1-4) and Commentary on Key Match Observations. Description of Key Situations and Their Resolution, Comprehensive Verbal Assessment of Referees as a Pair and Individually				
	R1	R2		
Strengths				
Weaknesses				
Video Materials from Match				
Situation 1				
Situation Type	Check	Situation Description		
Body Contact				
Stick Offence				
Technical Offence				
Unsporting Behaviour				
Other				
SharePoint Video Link:				
Sporfia Video Link:		PIN:		
Situation 2				
Situation Type	Check	Situation Description		
Body Contact				
Stick Offence				
Technical Offence				
Unsporting Behaviour				
Other				
SharePoint Video Link:				
Sporfia Video Link:		PIN:		
Situation 3				
Situation Type	Check	Situation Description		
Body Contact				
Stick Offence				
Technical Offence				
Unsporting Behaviour				

Other		
SharePoint Video Link:		
Sporfia Video Link:		PIN:

Notes		
Match Difficulty (Compared to Other Matches of the Same Competition)		Match Difficulty: Match difficulty considers the challenge of the match itself compared to other matches in the same competition. It reflects the style of play presented by teams and how often referees were required to intervene. It considers pressure on referees from players, team staff and spectators. Match progress and score development can also often influence difficulty.
4	VERY DIFFICULT	
3	DIFFICULT	
2	STANDARD MATCH FOR THIS COMPETITION	
1	EASY	
Referee Self-Assessment		Self-Assessment: Self-assessment considers how the referees themselves perceived the match and to what extent this perception aligns with the referee coach's perception and feedback.
4	Completely aligns with referee coach's view	
3	Mostly aligns with referee coach's view	
2	Mostly differs from referee coach's view on key points	
1	Completely differs from referee coach's view on key points	
Tolerance Level		Tolerance Level: Tolerance level considers how much referees intervened in the match and how strictly they assessed challenges (stick and body). Grade 1 means overly strict assessment delayed the match or clean challenges were penalized. Grade 4 means referees called very little - mostly just out balls and goals.
4	VERY LENIENT (referees called almost no offences)	
3	MORE LENIENT (referees tended to let the game flow)	
2	STRICTER (referees were rather strict)	
1	VERY STRICT (referees often called even "clean" challenges)	
Grade		Grade: The grade reflects how referees handled the match. Match difficulty is not factored into the grade. Even in an easy match, referees can receive grade 1 for excellent performance.
1	EXCELLENT (referee performance significantly exceeded match level)	
2	VERY GOOD (referees handled the match level without difficulty)	
3	SUFFICIENT (referees slightly fell short of match level)	
4	INSUFFICIENT (referees significantly fell short of match level)	

For the purpose of ranking referees by referee coach evaluations, each competition is assigned a difficulty coefficient. An excellent performance in a standard 8XMH match therefore has higher value than an excellent performance in a standard 4XE1 match. The coefficient table is available on the website upon request.

Figure B.4: The Map of the Selected Regions



Note: Red = Hradec Králové and Pardubice Region